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Securitization without risk transfer

Contents

1	Big picture	3
2	Data and measurement (ABCP conduits, guarantees, bank capital, and the run)	3
2.1	What an ABCP conduit is	3
2.2	Guarantee types	3
2.3	Conduit and sponsor data	4
2.4	Bank capital measures	4
2.5	ABCP run and rollover outcomes	4
2.6	Bank stock returns and missing capital	5
3	Models and empirical specifications (all equations plus interpretation)	5
3.1	Regulatory-arbitrage mechanism	5
3.2	Bank capital and conduit exposure	5
3.3	Guarantees during the ABCP run	5
3.4	Ex-post risk transfer	6
3.5	Conduit exposure and bank stock returns	6
3.6	Missing capital calculation	6
4	Identification and empirical design (what makes the estimates credible)	6
4.1	What identifies the regulatory-arbitrage result	6
4.2	Why the guarantee comparison matters	7
4.3	Why August 9, 2007 matters	7
4.4	Why realized losses matter	7
4.5	What the paper cannot fully prove	7
5	Tables and figures	8
5.1	Figures 1-2: the ABCP boom, run, and capital regulation	8
5.2	Tables 1-2: conduit scale, sponsors, guarantees, and assets	8
5.3	Figure 3 and Table 3: sponsor types and bank exposure	9
5.4	Tables 4-5: bank capital and conduit exposure	10
5.5	Tables 6-7: guarantee strength during the run	10
5.6	Tables 8-10: realized losses, bank stock returns, and missing capital	11
6	Short summary	12
7	Conclusion	13
7.1	Core conclusion	13

7.2	Economic intuition	13
7.3	Takeaway	13

1 Big picture

- **Question.** Did asset-backed commercial paper (ABCP) conduits transfer securitized risks away from banks, or did they let banks keep the risk while lowering regulatory capital?
- **Main setting.** The paper studies ABCP conduits, off-balance-sheet special purpose vehicles that bought medium- and long-term assets and funded them with short-term ABCP.
- **Core claim.** The conduits were securitization without meaningful risk transfer. Banks sponsored vehicles, gave investors explicit guarantees, and therefore kept the downside risk.
- **Central mechanism.** Liquidity guarantees were treated favorably for regulatory capital, but in practice were structured so that banks would repay ABCP investors in almost all relevant states.
- **Scale.** ABCP outstanding reached roughly \$1.3 trillion in July 2007, making it one of the largest money market instruments before the crisis.
- **Run.** After the BNP Paribas announcement on August 9, 2007, ABCP spreads jumped and outstanding fell sharply. This was a shadow-bank run.
- **Regulatory-arbitrage evidence.** More economically capital-constrained banks were more exposed to liquidity-guaranteed conduits, but not to credit-guaranteed conduits that did not provide similar capital relief.
- **Guarantee evidence.** During the run, conduits with weak guarantees, especially extendible notes and SIVs, had more difficulty rolling over ABCP and paid higher spreads. Liquidity-guaranteed conduits behaved more like credit-guaranteed conduits.
- **Realized losses.** Investors in liquidity-guaranteed and credit-guaranteed ABCP were repaid in full. Losses remained mostly with sponsoring banks.
- **Stock-market evidence.** Banks with more conduit exposure had lower stock returns around the start of the ABCP run.
- **Economic takeaway.** Shadow banking can look like risk transfer while actually concentrating risk inside regulated banks. Capital rules that focus on legal form rather than economic substance can generate systemic risk.

2 Data and measurement (ABCP conduits, guarantees, bank capital, and the run)

2.1 What an ABCP conduit is

- A conduit is a special purpose vehicle managed by a sponsoring financial institution.
- It buys assets such as mortgages, receivables, loans, and asset-backed securities.
- It finances these assets mostly by issuing very short-term ABCP.
- The conduit therefore performs maturity transformation: it holds long assets and issues short liabilities.
- Investors in ABCP are mainly money market funds and similar investors seeking safe short-term paper.

Interpretation. Conduits are economically bank-like even if legally off balance sheet. The key question is whether risk really leaves the bank.

2.2 Guarantee types

The paper classifies guarantees from strongest to weakest:

- **Credit guarantees:** the sponsor promises to pay ABCP investors regardless of conduit asset values. These were treated similarly to on-balance-sheet financing for regulatory capital.
- **Liquidity guarantees:** the sponsor pays if the conduit cannot roll over ABCP, except in formally defined asset-default states. In practice, default triggers were slow-moving, so these guarantees covered most economically relevant risks.
- **Extendible notes:** the conduit can extend the maturity of maturing paper. This delays payment and increases investor exposure to asset deterioration.
- **Structured investment vehicles (SIVs):** the conduit issues uninsured longer-maturity debt in addition to ABCP, making the guarantee weaker.

Interpretation. The regulatory distinction between credit and liquidity guarantees is central. Liquidity guarantees looked weaker in legal form, but were often strong enough to protect ABCP investors.

2.3 Conduit and sponsor data

- The conduit universe comes from Moody's ratings data from January 2001 to December 2009.
- The paper augments Moody's data with Bankscope information on sponsors and with ratings reports for asset types and guarantee structures.
- As of January 1, 2007, the data include 301 conduits with \$1,236.2 billion in ABCP.
- Commercial banks sponsored \$911.4 billion, around 74 percent of the total.
- Most top conduits held highly rated assets and used liquidity guarantees.

Interpretation. The data are close to the universe of rated ABCP conduits. This lets the authors study both structure before the crisis and outcomes during the run.

2.4 Bank capital measures

The paper distinguishes two capital ratios.

- **Economic capital:** book equity divided by assets, called the leverage ratio in the paper.
- **Regulatory capital:** Tier 1 regulatory capital divided by risk-weighted assets.

Interpretation. Regulatory arbitrage predicts a stronger relation with economic capital than with regulatory capital, because the purpose of conduits is to make regulatory capital look stable while economic leverage rises.

2.5 ABCP run and rollover outcomes

- The main run date is August 9, 2007, when BNP Paribas halted withdrawals from funds exposed to mortgage-backed securities.
- The paper studies weekly ABCP outstanding and daily overnight ABCP spreads before and after this event.
- It compares conduits with different guarantees.
- It also uses Moody's announcements to determine whether conduits remained active, repaid investors, or defaulted.

Interpretation. If guarantees transferred risk, stronger and weaker guarantee types should not differ much during the run. If liquidity guarantees kept risk with banks, liquidity-guaranteed ABCP should remain safer for investors than weakly guaranteed ABCP.

2.6 Bank stock returns and missing capital

- Bank stock returns are measured around the start of the ABCP run.
- Conduit exposure is measured as ABCP exposure divided by bank equity.
- Missing capital is calculated as the capital a bank would have needed if ABCP assets had required an 8 percent capital charge.

Interpretation. The stock-return test asks whether investors understood that the ABCP run was a bank-balance-sheet problem, not merely a conduit-investor problem.

3 Models and empirical specifications (all equations plus interpretation)

3.1 Regulatory-arbitrage mechanism

The paper’s main mechanism can be written as

Capital rule gap $\Rightarrow$ off-balance-sheet conduit with liquidity guarantee $\Rightarrow$ lower reported regulatory capital need $\Rightarrow$ risk recomb

Interpretation. The conduit is valuable to the bank because it lowers required capital, not because it genuinely transfers risk to investors.

3.2 Bank capital and conduit exposure

The baseline bank-level panel regression is

$$Exposure_{it} = \alpha_i + \delta_t + \beta CapitalRatio_{it} + \gamma' X_{it} + \varepsilon_{it}. \quad (1)$$

- $Exposure_{it}$ is ABCP exposure relative to bank equity.
- $CapitalRatio_{it}$ is either the Tier 1 ratio or the leverage ratio.
- X_{it} includes assets, return on assets, short-term debt share, deposit share, and loan share.
- α_i are bank fixed effects and δ_t are year fixed effects.

Interpretation. A negative β means banks with lower capital ratios set up more conduits. The key result is that this is strong for the leverage ratio and liquidity guarantees, but not for credit guarantees.

3.3 Guarantees during the ABCP run

The conduit-level run regression is

$$y_{it} = \alpha + \sum_j \beta_j Guarantee_{ij} + \sum_j \gamma_j (After_t \times Guarantee_{ij}) + \delta After_t + \varepsilon_{it}. \quad (2)$$

- y_{it} is either log ABCP outstanding or the overnight ABCP spread.
- $Guarantee_{ij}$ indicates guarantee type j .
- Liquidity guarantees are the omitted guarantee category.
- $After_t$ equals one after August 9, 2007.

Interpretation. Negative interaction coefficients for weak guarantees in the outstanding regression mean those conduits could not roll over paper. Positive interaction coefficients in the spread regression mean investors demanded compensation for weaker protection.

3.4 Ex-post risk transfer

The risk-transfer test is descriptive but powerful. The paper classifies each conduit after the crisis as

Active, Repaid, In default.

It then calculates estimated losses under assumed loss rates of 5 percent and 15 percent:

$$EstimatedLoss = ABCP \times LossRate, \quad (3)$$

allocated between sponsors and investors based on whether investors were repaid.

Interpretation. If securitization transferred risk, outside investors should bear substantial losses. Instead, investors protected by liquidity or credit guarantees were repaid, so losses remained with sponsors.

3.5 Conduit exposure and bank stock returns

The event-study regression is

$$R_i = \alpha + \beta ConduitExposure_i + \gamma' X_i + \varepsilon_i. \quad (4)$$

- R_i is bank i 's cumulative equity return over the event window.
- The baseline event window is August 8-10, 2007.
- $ConduitExposure_i$ is ABCP exposure relative to bank equity.
- Controls include assets, return on assets, short-term debt share, deposit share, loan share, and country effects.

Interpretation. If investors recognize that conduits create recourse to banks, higher exposed banks should lose market value when the run begins.

3.6 Missing capital calculation

For each sponsor, the missing capital calculation is

$$MissingCapital_i = 0.08 \times ABCP_i, \quad (5)$$

with the percent missing capital measured as

$$MissingCapitalShare_i = \frac{0.08 \times ABCP_i}{Tier1Capital_i}. \quad (6)$$

Interpretation. The calculation asks how much capital banks avoided holding by putting assets into conduits rather than keeping them on balance sheet.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the regulatory-arbitrage result

- The main identifying variation is within-bank and across-time variation in bank capital ratios and conduit exposure.
- The dependent variable is separated by guarantee type.

- The key comparison is liquidity-guaranteed ABCP versus credit-guaranteed ABCP.
- Credit guarantees are similar economically but do not provide the same capital relief.

Interpretation. If the same banks use liquidity conduits when economically capital constrained, but not credit conduits, the evidence points to regulatory capital relief rather than generic conduit demand.

4.2 Why the guarantee comparison matters

- Liquidity and credit guarantees both protected ABCP investors strongly during the run.
- But they differed sharply in regulatory treatment before the crisis.
- Extendible and SIV guarantees were weaker and therefore useful as a benchmark during the run.

Interpretation. The design exploits a gap between legal/regulatory classification and economic risk exposure.

4.3 Why August 9, 2007 matters

- August 9 is the start of the ABCP run in the paper's design.
- ABCP spreads and rollovers moved sharply after the BNP Paribas announcement.
- The event helps separate conduit-specific exposure from slower-moving bank characteristics.

Interpretation. The date is not a perfect randomized shock, but it is a sharp information event that revealed the fragility of ABCP financing.

4.4 Why realized losses matter

- The ultimate risk-transfer test is who lost money.
- The paper hand-collects Moody's announcements on conduit defaults and repayments.
- Liquidity and credit guarantee investors were fully repaid, while weak-guarantee investors suffered some losses.

Interpretation. Legal structure and ex-post repayment outcomes tell the same story: most risk returned to sponsors.

4.5 What the paper cannot fully prove

- The decision to sponsor conduits was not randomly assigned.
- Banks may have had unobserved business models correlated with both capital ratios and conduit use.
- Actual asset losses are estimated because detailed conduit asset-level loss data are not fully public.
- The stock-return event study captures market perceptions and may be affected by other simultaneous crisis news.

Interpretation. The paper is strongest because several independent tests all point in the same direction: regulatory arbitrage, weak risk transfer, sponsor losses, and bank stock-market exposure.

5 Tables and figures

5.1 Figures 1-2: the ABCP boom, run, and capital regulation

Read:

- Figure 1 shows ABCP outstanding rising toward \$1.3 trillion before the run, then falling sharply after August 2007.
- The overnight ABCP spread over the federal funds rate jumps dramatically after August 9, 2007.
- Figure 2 shows ABCP growth stalling when regulators considered tougher treatment after Enron and FASB consolidation discussions.
- ABCP growth accelerates after regulators provided a permanent exemption in 2004.

Economic message: The timing is consistent with capital regulation shaping the size of the conduit sector. When capital treatment became more favorable, the sector expanded rapidly.

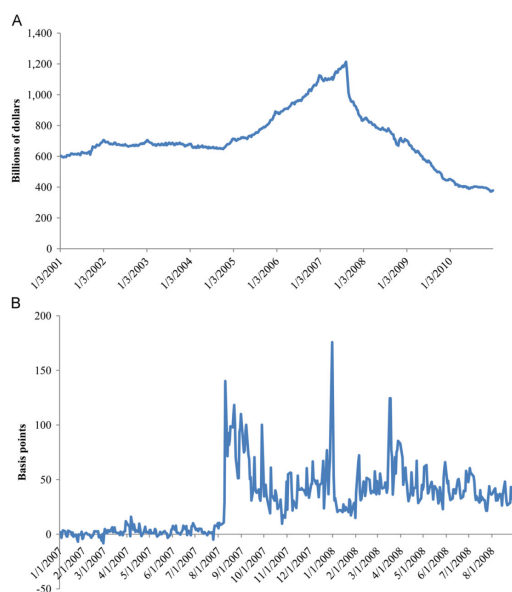
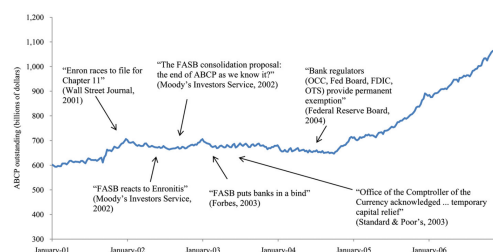


Fig. 1. Asset-backed commercial paper (ABCP) outstanding and spreads. Panel A plots weekly total ABCP outstanding in the US market from January 2001 to April 2010. Panel B shows the daily spread of overnight ABCP over the federal funds rate from January 2007 to August 2008. The figures are based on data published by the Federal Reserve Board.



2. Asset-backed commercial paper (ABCP) outstanding and capital regulation. This figure shows total weekly ABCP outstanding from January 2001 to December 2006. The figure also shows the timeline regulatory decisions on regulatory capital required for guarantees provided to conduits.

5.2 Tables 1-2: conduit scale, sponsors, guarantees, and assets

Read:

- Table 1 shows that large conduits were typically liquidity-guaranteed and sponsored by major commercial banks.
- Citigroup sponsored \$92.7 billion of ABCP, roughly comparable to its \$90.9 billion of Tier 1 capital.
- ABN Amro sponsored \$68.6 billion against \$31.2 billion of Tier 1 capital.
- Table 2 shows 301 conduits with \$1,236.2 billion in ABCP as of January 2007.
- Liquidity guarantees covered \$752.9 billion, the largest guarantee class.
- Commercial banks sponsored \$911.4 billion, far more than structured finance firms or mortgage lenders.

Economic message: The conduit system was not peripheral. It was large, bank-sponsored, and mostly backed by guarantees that looked like liquidity support but often functioned like credit support.

Table 1

Conduits and sponsors.

This table shows the ten largest conduits and sponsors as of January 1, 2007. The sample is restricted to bank-sponsored conduits. The information is collected from Moody's Rating Reports and BankScope. "ABCP (billion)" denotes asset-backed commercial paper (ABCP) outstanding per conduit (Panel A) and sponsor (Panel B). "Asset origin," "Asset rating," and "Asset type" denote characteristics of the main asset class owned by a conduit; CDO/CLO represents combinations of collateralized debt obligations and collateralized loan obligations.

Panel A: Ten largest conduits						
Program name	Sponsor	ABCP (billion)	Guarantee	Asset origin	Asset rating	Asset type (share)
Grampian Funding	HBOS	37.9	Liquidity	United States	AAA	Residential mortgages (36%)
Amstel Funding	ABN Amro	30.7	Liquidity	Netherlands	AAA	CDO/CLO (84%)
Scaldis Capital	Fortis Bank	22.6	Liquidity	United States	AAA	Asset-backed securities (77%)
Sheffield Receivables	Barclays	21.4	Liquidity	n.a.	NR	Mortgages (43%)
Morrigan TRR	Hypo Public	18.9	Credit	n.a.	n.a.	Bonds (51%)
Cancara Asset	Lloyds	18.8	Liquidity	Great Britain	AAA	Residential mortgages (43%)
Solitaire Funding	HSBC	18.5	Liquidity	United States	AAA	Residential mortgages (45%)
Rhineland Funding	IKB	16.7	Liquidity	United States	AAA	CDO/CLO (95%)
Mane Funding	ING	13.7	Liquidity	n.a.	AAA	Asset-backed securities (91%)
Atlantis One	Rabobank	13.5	Liquidity	United States	NR	Commercial loans (100%)
Panel B: Ten largest sponsors						
Sponsor	Country	ABCP (billion)	Assets (billion)	Tier 1 capital (billion)	ABCP/Tier 1 (percent)	Tier 1 ratio (percent)
Citigroup	United States	92.7	1,884.3	90.9	102.0	8.6
ABN Amro	Netherlands	68.6	1,300.0	31.2	219.5	8.5
Bank of America	United States	45.7	1,459.7	91.1	50.2	8.6
HBOS Plc	Great Britain	43.9	1,161.7	44.0	99.7	8.1
J.P. Morgan	United States	42.7	1,351.5	81.1	52.7	8.7
HSBC	Great Britain	39.4	1,860.8	87.8	44.9	9.4
Deutsche Bank AG	Germany	38.7	2,070.0	31.0	125.0	8.5
Société Générale	France	38.6	1,260.2	29.4	131.3	7.8
Barclays Plc	Great Britain	33.1	1,956.7	45.2	73.2	7.7
Rabobank	Netherlands	30.7	732.9	34.8	88.3	10.7

Table 2

Conduit and sponsor statistics.

This table includes all conduits rated by Moody's Investors Service as of January 1, 2007. Panel A shows summary statistics by conduit. "Risk transfer" refers to the sponsor guarantee. "Assets" is the conduit's main asset type. "Currency" is the conduit's issuing currency. Panel B aggregates conduits by sponsor. "Sponsor type" is the sponsor type. "Country of origin" denotes the sponsor's headquarters.

Panel A: Conduits				
	Total		Size	
	Number of conduits	Size (billion)	Mean	Standard deviation
All conduits	301	1,236.2	4.1	(5.1)
Risk transfer				
Liquidity	163	752.9	4.6	(5.7)
Credit	55	199.9	2.9	(4.6)
Extendible notes	55	230.9	4.2	(4.5)
Structured investment vehicle	28	92.6	3.3	(3.4)
Assets				
Asset-backed securities	91	387.4	4.2	(5.9)
Loans	39	65.3	1.6	(2.4)
Receivables	88	436.7	3.5	(4.9)
Mixed asset categories	59	272.9	4.6	(5.3)
Other	24	74.0	4.9	(4.7)
Currency				
US dollar	233	973.0	4.2	(4.6)
Euro	33	220.0	6.7	(8.4)
Other	35	43.2	1.2	(1.6)
Panel B: Sponsors				
	Total		Size	
	Number of sponsors	Size (billion)	Mean	Standard deviation
All programs	127	1,236.2	9.7	(14.7)
Sponsor type				
Commercial banks	67	911.4	13.6	(17.6)
Structured finance	19	155.8	8.2	(13.7)
Mortgage lender	18	75.5	4.2	(5.8)
Investment manager	5	17.6	3.5	(3.3)
Investment banks	4	11.0	2.7	(2.2)
Other	14	94.8	4.6	(6.2)
Country of origin				
United States	67	491.8	7.3	(14.7)
Germany	15	204.1	13.6	(11.6)
United Kingdom	10	195.7	19.6	(17.0)
Other	35	344.5	9.8	(14.4)

5.3 Figure 3 and Table 3: sponsor types and bank exposure

Read:

- Figure 3 shows that commercial banks were the dominant sponsors of liquidity-guaranteed ABCP.
- Structured finance companies and mortgage originators relied more heavily on weaker guarantees.
- Table 3 reports mean total conduit exposure of 32.2 percent of equity and liquidity-guaranteed exposure of 25.2 percent of equity.
- Table 3 also shows a negative correlation between leverage ratio and conduit exposure, especially for liquidity-guaranteed exposure.

Economic message: The institutions most constrained by bank capital rules - commercial banks - used the capital-relieving guarantee category most heavily.

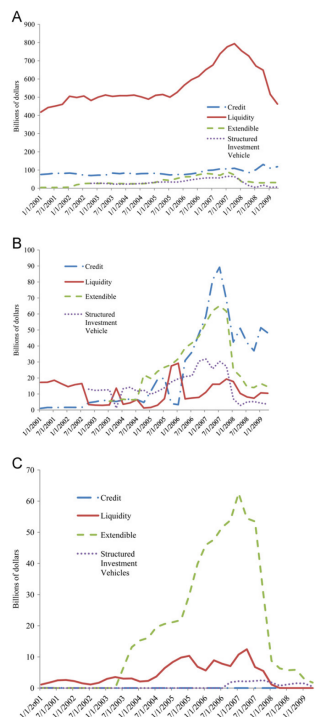


Fig. 3. Asset-backed commercial paper (ABCP) outstanding by sponsor and guarantee. This figure shows quarterly ABCP outstanding by guarantee and sponsor type from January 2001 to April 2009. Panel A plots commercial banks, Panel B shows structured finance companies, and Panel C shows mortgage originators. The figures are based on data provided by Moody's Investors Service.

Figure 3: ABCP by sponsor and guarantee

Table 3
Commercial banks and conduit activity.
This table shows conduit exposure of commercial banks. Panel A provides summary statistics for commercial banks with more than \$50 billion in assets headquartered in Europe or the United States for the fiscal years 2000-2006. "Conduit exposure" is the ratio of asset-backed commercial paper outstanding to total equity. "Leverage ratio" is the ratio of equity to assets. "Tier 1 ratio" is the ratio of Tier 1 to risk-weighted assets. "Assets" and "Log(Assets)" are total assets and the logarithm of total assets, respectively. "Return on assets" is the ratio of net profit to assets. "Share short-term debt," "Share deposits," and "Share loans" are short-term debt, banks deposits, and loans as a share of total assets, respectively. Panel B provides correlations between the main variables.

Panel A: Summary statistics (126 banks)							
Variable	Mean	Standard deviation	Median	Minimum	Maximum	N	
Conduit exposure (total)	32.20%	81.50%	0.00%	0.00%	999.10%	814	
Conduit exposure (liquidity)	25.20%	63.20%	0.00%	0.00%	726.30%	814	
Conduit exposure (credit)	1.80%	8.10%	0.00%	0.00%	89.70%	814	
Leverage ratio	5.40%	2.60%	4.90%	0.80%	16.90%	814	
Tier 1 ratio	8.41%	2.00%	8.00%	4.30%	19.00%	814	
Assets (billions of dollars)	260.8	326.4	134.9	9.7	2,070	814	
Log(Assets)	4.973	1.086	4.90%	2.27	7.635	814	
Return on assets	0.80%	0.60%	0.70%	-1.70%	3.10%	814	
Share short-term debt	11.70%	9.90%	9.50%	0.00%	51.60%	814	
Share deposits	57.60%	13.50%	59.80%	1.80%	86.80%	814	
Share loans	54.20%	17.10%	55.90%	4.80%	85.90%	814	
Panel B: Correlations (N=814)							
Variable	Conduit exposure (total)	Conduit exposure (liquidity)	Conduit exposure (credit)	Leverage ratio	Tier 1 ratio	Log Assets	Return on assets
Conduit exposure (total)	1						
Conduit exposure (liquidity)	0.7929	1					
Conduit exposure (credit)	0.2009	0.1445	1				
Leverage ratio	-0.2013	-0.1969	0.0119	1			
Tier 1 ratio	-0.0318	-0.035	-0.0265	0.2582	1		

Table 3: bank exposure summary

5.4 Tables 4-5: bank capital and conduit exposure

Read:

- Table 4 shows that Tier 1 ratios are negatively but weakly related to liquidity conduit exposure.
- The leverage ratio is strongly negatively related to liquidity conduit exposure across specifications.
- A one-standard-deviation increase in the leverage ratio reduces liquidity conduit activity by about 10.4 percent, about 40 percent of mean liquidity exposure.
- Table 5 repeats the same tests for credit-guaranteed exposure.
- The coefficients are statistically insignificant and close to zero.

Economic message: The regulatory-arbitrage interpretation is strong because capital-constrained banks expand capital-relieving liquidity conduits, not non-capital-relieving credit conduits.

5.5 Tables 6-7: guarantee strength during the run

Read:

- Table 6 compares conduit outcomes before and after August 9, 2007.
- Extendible notes and SIVs have large negative post-crisis coefficients for log outstanding.
- Extendible notes and SIVs also have positive post-crisis coefficients for ABCP spreads.
- Credit guarantees look similar to liquidity guarantees, the omitted category.
- Table 7 adds controls for asset categories and sponsor types and reaches similar conclusions.

Economic message: During the run, investors treated liquidity guarantees as strong guarantees. This supports the claim that liquidity guarantees did not actually transfer much risk away from sponsoring banks.

Table 4

Bank capital and conduit exposure (liquidity).

This table analyzes the relation between bank capital and exposure to conduits sponsored with liquidity guarantees. The sample includes commercial banks with more than \$50 billion in assets based in Europe and the United States in the fiscal years 2000–2006. The dependent variable “Conduit exposure (liquidity)” is total outstanding asset-backed commercial paper supported with liquidity guarantees relative to bank equity. In Panel A, the main independent variable is the ratio of Tier 1 capital to risk-weighted assets (“Tier 1 ratio”). In Panel B, the main independent variable is the ratio of book equity to assets (“Leverage ratio”). All regressions include controls for total assets, return on assets, short-term debt share, loan share, deposit share, and year fixed effects. Columns 1, 2, 4, and 5 include bank fixed effects, and Columns 2 to 5 include interactions of country-year fixed effects. Column 4 is estimated in first differences. Column 5 is estimated in first differences with a one-year lag. Standard errors reported in parentheses are clustered at the bank level (126 banks). ***, **, and * represent 1%, 5%, and 10% significance, respectively.

Panel A: Conduit exposure (liquidity)					
Estimation	Fixed effects (1)	Fixed effects (2)	Ordinary least squares (3)	First differences (4)	First differences-lagged (5)
Tier 1 ratio	−1.565 (1.225)	−0.578 (1.178)	−0.016 (2.308)	−0.183 (1.187)	−0.179 (1.160)
Log(Assets)	0.093 (0.075)	0.120 (0.095)	0.094* (0.048)	−0.128 (0.100)	0.034** (0.017)
Bank fixed effects	Yes	Yes	No	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	Yes	Yes	Yes	Yes
Number of banks	126	126	126	126	126
Number of observations	814	814	814	687	564
R-squared	0.850	0.868	0.295	0.145	0.130
Panel B: Conduit exposure (liquidity)					
Estimation	Fixed effects (1)	Fixed effects (2)	Ordinary least squares (3)	Fixed effects (4)	First differences-lagged (5)
Leverage ratio	−4.024** (1.903)	−4.625** (1.944)	−5.874** (2.709)	−4.870** (1.389)	−2.452** (0.970)
Log(Assets)	0.095 (0.075)	0.120 (0.093)	0.095* (0.048)	−0.124 (0.104)	−0.080 (0.067)
Bank fixed effects	Yes	Yes	No	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	Yes	Yes	Yes	Yes
Number of banks	126	126	126	126	126
Number of observations	814	814	814	687	564
R-squared	0.850	0.868	0.295	0.145	0.130

Table 6

Effect of guarantee on asset-backed commercial paper (ABCP) outstanding and spreads.

This table shows the effect of guarantees on paper outstanding and spreads from April to December 2007. The dependent variable in Columns 1 to 4 is the weekly log of paper outstanding and in Columns 5 to 8 is the daily overnight spread over the federal funds rate. “Credit,” “Extendible,” and “SIV” (structured investment vehicle) are indicators for the type of guarantee. “After” is an indicator for dates after August 9, 2007. Columns 2, 3, 6 and 7 include time fixed effects. Columns 3, 4, 7, and 8 include conduit fixed effects. Columns 4 and 8 include sponsor-time fixed effects. Standard errors in parentheses are clustered at the conduit level. ***, **, and * represent 1%, 5%, and 10% significance, respectively.

	Log(Outstanding)				Spread			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CreditsAfter	−0.068 (0.124)	−0.062 (0.125)	0.061 (0.130)	−0.005 (0.202)	0.023 (0.055)	0.025 (0.055)	0.041 (0.058)	−0.004 (0.103)
ExtendibleAfter	−0.725** (0.291)	−0.748** (0.284)	−0.880** (0.200)	−0.081* (0.404)	0.129** (0.054)	0.093** (0.047)	0.135** (0.050)	0.068 (0.110)
SIVsAfter	−0.697** (0.156)	−0.684** (0.157)	−0.563** (0.157)	−0.454 (0.290)	0.316** (0.099)	0.254** (0.082)	0.260** (0.093)	0.315** (0.132)
Credit	−0.419 (0.376)	−0.419 (0.377)	−0.419 (0.377)	0.000 (0.005)	0.000 (0.005)	0.001 (0.005)	0.001 (0.005)	0.000 (0.005)
Extendible	0.132 (0.294)	0.132 (0.294)	0.132 (0.294)	0.022** (0.009)	0.022** (0.009)	0.022** (0.009)	0.022** (0.009)	0.022** (0.009)
SIV	−0.336** (0.167)	−0.336** (0.167)	−0.336** (0.167)	0.001 (0.006)	0.001 (0.006)	0.001 (0.006)	0.001 (0.006)	0.001 (0.006)
After	−0.211** (0.084)	−0.211** (0.084)	−0.211** (0.084)	0.474** (0.028)	0.474** (0.028)	0.474** (0.028)	0.474** (0.028)	0.474** (0.028)
Number of observations	7,630	7,630	7,630	7,630	14,862	14,862	14,862	14,862
R-squared	0.053	0.057	0.049	0.937	0.444	0.717	0.843	0.952
Time fixed effects	No	Yes	Yes	No	No	Yes	Yes	No
Sponsor-time fixed effects	No	No	No	Yes	No	No	No	Yes
Conduit fixed effects	No	No	Yes	Yes	No	No	Yes	Yes

Table 5

Bank capital and conduit exposure (credit).

This table analyzes the relation between bank capital and exposure to conduits sponsored with liquidity guarantees. The sample includes commercial banks with more than \$50 billion in assets based in Europe and the United States in the fiscal years 2000–2006. The dependent variable “Conduit exposure (liquidity)” is total outstanding asset-backed commercial paper supported with liquidity guarantees relative to bank equity. In Panel A, the main independent variable is the ratio of Tier 1 capital to risk-weighted assets (“Tier 1 ratio”). In Panel B, the main independent variable is the ratio of book equity to assets (“Leverage ratio”). All regressions include controls for total assets, return on assets, short-term debt share, loan share, deposit share, and year fixed effects. Columns 1, 2, 4, and 5 include bank fixed effects, and Columns 2 to 5 include interactions of country-year fixed effects. Column 4 is estimated in first differences. Column 5 is estimated in first differences with a one-year lag. Standard errors reported in parentheses are clustered at the bank level (126 banks). ***, **, and * represent 1%, 5%, and 10% significance, respectively.

Panel A: Conduit exposure (credit)					
Estimation	Fixed effects (1)	Fixed effects (2)	Ordinary least squares (3)	First differences (4)	First differences-lagged (5)
Tier 1 ratio	0.473 (0.411)	0.349 (0.416)	0.082 (0.489)	−0.139 (0.207)	−0.199 (0.160)
Log(Assets)	0.013 (0.024)	0.009 (0.031)	0.025** (0.011)	0.018 (0.025)	0.030* (0.015)
Bank fixed effects	Yes	Yes	No	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	Yes	Yes	Yes	Yes
Number of banks	126	126	126	126	126
Number of observations	814	814	814	687	564
R-squared	0.850	0.868	0.295	0.145	0.130
Panel B: Conduit exposure (credit)					
Estimation	Fixed effects (1)	Fixed effects (2)	Ordinary least squares (3)	First differences (4)	First differences-lagged (5)
Leverage ratio	−0.933 (0.591)	−0.834 (0.652)	0.145 (0.464)	−0.653 (0.402)	−0.179 (0.160)
Log(Assets)	0.003 (0.024)	0.003 (0.030)	0.024** (0.009)	0.021 (0.024)	0.034** (0.017)
Bank fixed effects	Yes	Yes	No	Yes	Yes
Bank controls	Yes	Yes	Yes	Yes	Yes
Country-year fixed effects	No	Yes	Yes	Yes	Yes
Number of banks	126	126	126	126	126
Number of observations	814	814	814	687	564
R-squared	0.679	0.733	0.222	0.228	0.250

Table 7

Effect of guarantee on asset-backed commercial paper (ABCP) outstanding and spreads (robustness).

This table shows the effect of guarantees on paper outstanding and spreads from April to December 2007. The dependent variable in Columns 1–4 is the weekly log of paper outstanding and in Columns 5–8 is the daily overnight spread over the federal funds rate. “Credit,” “Extendible,” and “SIV” (structured investment vehicle) are indicators for the type of credit guarantee. “After” is an indicator for dates after August 9, 2007. Columns 1, 3, 4, 5, 7, and 8 include time fixed effects. Columns 1, 2, 4, 5, 6 and 8 include conduit fixed effects. Columns 2 and 6 include sponsor-time fixed effects. Standard errors in parentheses are clustered at the conduit level. All columns include controls for type of assets and type of sponsor and interaction of the controls with “After.” ***, **, and * represent 1%, 5%, and 10% significance, respectively.

	Log(Outstanding)				Spread			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
CreditsAfter	0.065 (0.135)	0.023 (0.212)	0.007 (0.145)	0.073 (0.141)	−0.015 (0.054)	−0.100 (0.077)	−0.039 (0.053)	−0.061 (0.048)
ExtendibleAfter	−0.818** (0.201)	−0.883* (0.404)	−0.621** (0.231)	−0.623** (0.229)	0.021 (0.061)	−0.119 (0.214)	−0.069 (0.064)	−0.089 (0.071)
SIVsAfter	−0.451** (0.176)	−0.391 (0.330)	−0.544** (0.179)	−0.451** (0.183)	0.166 (0.109)	0.245** (0.107)	0.171** (0.082)	0.134 (0.085)
ReceivablesAfter	0.311** (0.153)	0.212 (0.228)	0.370** (0.148)	0.324** (0.145)	−0.221** (0.074)	−0.150* (0.089)	−0.192** (0.060)	−0.228** (0.060)
LoansAfter	−0.384 (0.289)	−0.506* (0.278)	−0.253 (0.255)	0.039 (0.208)	0.066 (0.171)	0.450** (0.083)	0.158 (0.161)	0.063 (0.170)
BanksAfter	0.051 (0.231)	0.046 (0.240)	0.051 (0.240)	0.046 (0.240)	0.051 (0.240)	0.046 (0.240)	0.051 (0.240)	0.046 (0.240)
Number of observations	7,630	7,630	7,630	7,630	14,862	14,862	14,862	14,862
R-squared	0.853	0.938	0.189	0.859	0.865	0.96	0.772	0.869
Time fixed effects	Yes	No	Yes	Yes	Yes	No	Yes	Yes
Sponsor-time fixed effects	No	Yes	No	No	No	Yes	No	No
Conduit fixed effects	Yes	Yes	No	Yes	Yes	Yes	No	Yes

5.6 Tables 8-10: realized losses, bank stock returns, and missing capital

Read:

- Table 8 shows that only 2.5 percent of pre-crisis ABCP entered default, while 76.6 percent remained active and 20.8 percent was repaid.
- Liquidity and credit guarantees had zero investor default in the table.
- Estimated losses under 5 percent and 15 percent loss rates are \$68.0 billion and \$204.1 billion for sponsors, versus only \$1.7 billion and \$5.2 billion for investors.
- Table 9 shows that a move from zero to 100 percent conduit exposure reduces the August 8-10 bank stock return by 1.096 percentage points with controls.
- The monthly August coefficient is -2.334 percentage points; placebo months before August are insignificant.
- Table 10 shows a total missing-capital calculation of \$68.9 billion for the 30 largest bank sponsors, equal to 6.1 percent of Tier 1 capital on average.

Economic message: These tests connect the contractual structure to real losses. Investors were mostly protected; banks bore the risk; and the stock market penalized exposed banks when the run began.

Table 9
Estimated losses for sponsors and outside investors.
The table shows three post-crisis risk transfer by month guarantee. "Pre-crisis" denotes total asset-backed commercial paper (ABCP) outstanding as of July 1, 2007. Post-crisis denotes the value-weighted share that is "Active" (conduit closed and equal investors), and "In default" (conduit closed and investors were not repaid). "Estimated loss" denotes the losses of sponsor and outside investors assuming a recovery rate on conduit assets of 95% and 85%, respectively.

	Pre-crisis				Post-crisis				Estimated loss (\$billions)	
	ABCP (\$billions)	Active	Repaid	In default	Sponsor	Investor	Sponsor	Investor	Loss rate 95%	Loss rate 85%
All	1,395.5	76.8	20.8	2.5%	68.0	1.7	294.1	5.2		
Guarantee					0.0	0.0	0.0	0.0		
Liquidity	844	87.0	12.1	0.0%	42.2	0.0	126.6	0.0		
Credit	264.2	76.0	20.1	0.0%	16.2	0.0	16.6	0.0		
Extendible	243.1	47.0	45.5	7.4%	11.3	0.0	31.8	2.7		
Investment investment vehicle	104.1	10.7	17.7	16.0%	4.3	0.0	11.0	2.6		
Vehicle type					0.0	0.0	0.0	0.0		
Commercial bank	1,031.6	81.0	16.4	0.0%	51.5	0.1	154.4	0.9		
Commercial finance	199.2	19.1	16.4	5.2%	4.4	0.1	16.2	1.6		
Mortgage lender	62.2	44.5	40.2	15.1%	15.3	2.5	0.5	7.0	1.4	
Other	106.4	61.3	24.4	8.0%	4.8	0.4	13.7	1.3		

Table 9
Event study - effect of conduit exposure on stock returns.
This table shows the effect of conduit exposure on stock returns. We restrict the sample to commercial banks that (i) have at least \$10 billion in assets (ii) are located in the Europe or the United States, and (iii) have share price data available. In Columns (1) and (2), the dependent variable is the total stock returns over the three-day period from August 9 to August 10, 2007. In Columns (3)-(10), the dependent variable is the stock returns in the month indicated under "Time". The measure "Conduit exposure (total)" is the ratio of equity claims to equity claims (2)-(10) minus sample bank effects. The control variables are defined in the Appendix. We report heteroskedasticity-consistent standard errors in parentheses. **, *, * represent 1%, 5%, and 10% significance, respectively.

Time	Dependent variable: Stock returns (percent)									
	Event regressions					Placebo regressions				
	August 9-10	August 9-10	August	January	February	March	April	May	June	July
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Conduit exposure	-1.842*** (0.402)	-1.296** (0.485)	-2.147* (1.096)	0.688 (1.288)	-0.275 (0.877)	0.122 (0.882)	-0.017 (0.872)	-0.348 (1.442)	4.401 (1.511)	1.983 (0.952)
Log(Size)	0.007 (1.197)	0.004 (1.020)	0.002 (1.086)	0.234 (1.054)	0.077 (0.811)	0.229 (0.869)	-1.058 (0.914)	1.215 (0.915)	0.262 (0.716)	0.062 (0.716)
Return on assets	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
Share short-term debt	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)	-1.072 (0.001)
Share deposits	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)	0.001 (0.001)
Share loans	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)	-0.001 (0.001)
Country fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Number of observations	88	88	88	88	88	88	88	88	88	88
R-squared	0.009	0.010	0.014	0.016	0.019	0.021	0.024	0.029	0.034	0.040

Table 10
Missing capital.
This table lists the 30 largest bank sponsors of asset-backed commercial paper (ABCP) as of January 1, 2007. For each bank, we compute the required capital assuming ABCP requires a capital charge of 8%, i.e., ABCP*0.08=Total, expressed in billions of US dollars. We also compute the missing capital as a share of the bank's equity. We measure equity as Tier 1 capital. If a bank does not report Tier 1 capital, we multiply shareholder equity with the average Tier 1/equity shareholder ratio of banks that report both shareholder equity and Tier 1 ratio.

Name	Missing capital			
	Tier 1	ABCP	Total	Percent
Citigroup	90.9	92.7	7.4	8.2
ABN Amro	31.2	68.6	5.5	17.6
Bank of America	91.1	45.7	3.7	4.0
HBOS	44	43.9	3.5	8.0
JP Morgan Chase	81.1	42.7	3.4	4.2
HSBC Holdings	87.8	39.4	3.2	3.6
Deutsche Bank	31	38.7	3.1	10.0
Société Générale	29.4	38.6	3.1	10.5
Barclays	45.2	33.1	2.6	5.9
Mitsubishi UFJ Financial Group	68.5	32.0	2.6	3.7
Rabobank	34.8	30.8	2.5	7.1
Westdeutsche Landesbank	9.5	29.9	2.4	25.1
ING Groep	54.3	26.4	2.1	3.9
Dresdner Bank	18.7	23.2	1.9	9.9
Fortis	16.4	22.6	1.8	11.0
Bayerische Landesbank	15.8	22.4	1.8	11.3
Bayerische Hypo-und Vereinsbank	14.1	22.3	1.8	12.6
State Street Corporation	24.1	21.9	1.7	7.2
Crédit Agricole	6.5	19.5	1.6	24.1
Hypo Real Estate	4.5	18.9	1.5	33.4
Lloyds Banking Group	6.1	18.8	1.5	24.6
Countrywide Financial Corporation	25.2	18.3	1.5	5.8
CMAC	15.4	17.5	1.4	9.1
Royal Bank of Scotland	75.2	15.8	1.3	1.7
Royal Bank of Canada	52.3	15.6	1.2	2.4
Bear Stearns Companies	19.1	13.8	1.1	5.8
KBC Group	22.9	12.6	1	4.4
Sachsen Landesbank	1.3	12.5	1	79.9
BNP Paribas	62.3	11.6	0.9	1.5
Bank of Montreal	45.3	11.5	0.9	2.0
Total	1,124.0	861.5	68.9	6.1

6 Short summary

- The paper studies ABCP conduits as a major shadow-banking structure before the 2007-2009 crisis.
- Conduits bought longer-term assets and issued short-term ABCP.
- Sponsors provided guarantees that made ABCP safe for outside investors.
- Liquidity guarantees received favorable regulatory capital treatment.
- In practice, liquidity guarantees were often strong enough to protect investors from credit and rollover losses.
- Commercial banks sponsored most of the liquidity-guaranteed ABCP.
- More economically capital-constrained banks sponsored more liquidity-guaranteed ABCP.
- The same capital relation does not appear for credit-guaranteed ABCP, which did not provide the same regulatory relief.
- The ABCP market grew rapidly after regulatory decisions in 2004 confirmed favorable treatment.
- After August 9, 2007, ABCP spreads spiked and outstanding fell sharply.
- Weakly guaranteed conduits, especially extendible notes and SIVs, had more trouble rolling over paper and paid higher spreads.
- Liquidity-guaranteed conduits performed more like credit-guaranteed conduits during the run.
- Almost all ABCP investors were repaid; losses were largely borne by sponsors.
- Banks with more conduit exposure had lower stock returns around the run.
- The evidence supports regulatory arbitrage rather than genuine risk transfer.
- The paper illustrates how shadow banking can remain tightly connected to regulated banks through explicit recourse.

7 Conclusion

7.1 Core conclusion

Acharya, Schnabl, and Suarez show that ABCP conduits were a form of securitization without meaningful risk transfer. Banks used off-balance-sheet vehicles to reduce regulatory capital requirements while providing guarantees that returned risk to their own balance sheets in crisis states.

7.2 Economic intuition

The economic intuition is that capital regulation created a wedge between legal form and economic exposure. If a bank could call its guarantee a liquidity guarantee, it could lower required capital. But if the guarantee was written so that investors were protected in almost all relevant states, then the bank still bore the economic risk. The conduit therefore delivered capital relief without true risk dispersion.

7.3 Takeaway

The paper's broader lesson is that securitization and shadow banking are not automatically risk-transfer mechanisms. What matters is the recourse structure. If investors are insured by banks, then the risk remains in the banking sector, even when assets appear to leave bank balance sheets.